

DESIGN THINKING AND INNOVATION

Digital Adoption Readiness Tool

A human-centred digital solution for small businesses

Semester 1 Project Report

Bachelor of Engineering — Computer Science and Engineering (AI & ML)

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Project link: digital-adoption-readiness-tool.netlify.app

1 Executive Summary

Small businesses are increasingly expected to accept digital payments, maintain an online presence, organise records, and communicate with customers through digital channels. However, many owners do not know which improvement should come first. The **Digital Adoption Readiness Tool** addresses this gap through a short, plain-language self-assessment that converts a business owner's answers into a practical, prioritised roadmap.

The tool was designed using the principles of design thinking: understanding the user's context, defining a focused problem, generating accessible ideas, prototyping a usable experience, and testing the interaction flow. It supports English, Hindi, and Punjabi so that language does not become an additional barrier to adoption. The final product combines a friendly interface, a rule-based scoring model, free tool recommendations, and persistent storage for completed check-ins.

Live website: [Open the Digital Adoption Readiness Tool](#)

Submission note: The public Netlify deployment contains the interactive frontend. The full database-backed version is maintained in the project environment for development and demonstration.

2 Problem Identification

Many local shops, service providers, and home businesses want to benefit from digital tools but face three practical difficulties. First, digital advice is often too broad or technical. Second, owners may not know which improvement will create the greatest immediate value. Third, a solution that ignores language, confidence, time, or daily business routines is unlikely to be used consistently.

The project therefore focuses on a simple design question:

“How might we help a small-business owner identify one realistic digital improvement and begin it without feeling overwhelmed?”

2.1 Target users

The primary users are owners or operators of small local businesses, including shops, home businesses, and service providers. They may have a smartphone and access to common digital services, but they may not have formal digital training. Their most important needs are clarity, trust, low effort, and an immediately useful next step.

3 Design Thinking Process

3.1 Empathise

The experience was shaped around the everyday reality of a small business owner. The interface avoids specialist vocabulary and asks about familiar activities: receiving payments, being found online, tracking sales and stock, staying in touch with customers, understanding money flow, and trying a new tool. The language and tone are deliberately respectful rather than judgemental.

3.2 Define

The defined need is not “more information about technology.” It is a clearer starting point. The tool must tell a user where they currently stand, identify the largest opportunity, and recommend a manageable action that can be attempted in a short period of time.

3.3 Ideate

Several product ideas were combined into one focused flow:

- a seven-question digital readiness check-in;
- four understandable readiness areas: payments, online presence, records, and marketing/customer connection;
- a percentage score that is presented as guidance rather than a grade;
- a roadmap of free-to-try tools;
- language switching between English, Hindi, and Punjabi; and
- a small, encouraging visual system that feels more like a helpful guide than an institutional form.

3.4 Prototype

The prototype was implemented as a responsive web tool. The landing screen introduces the purpose in one sentence, the quiz uses one question per screen, and the result page turns the score into a practical sequence of recommendations. The design uses a calm cream background, teal action colour, clear cards, and a storefront illustration to connect the digital concept with local business life.

3.5 Test and refine

The completed flow was tested from introduction to final roadmap in the browser. The project was also checked with TypeScript validation, a production build, and automated tests for authentication and readiness submission behaviour. A completed check-in was saved and verified in the database during development.

4 Solution Overview

The tool follows a simple three-step journey:

Step	User experience
1. Check in	The user answers seven short questions using familiar options such as “Always”, “Sometimes”, “Rarely”, and “Never”.
2. Understand	The tool calculates an overall readiness percentage and shows strengths and opportunities across four areas.
3. Act	The user receives a prioritised roadmap with free tools and a small first action to try.

4.1 Key features

Human-centred questions. Each question is written in plain language and includes a short explanation so the user can answer confidently.

Multilingual access. English, Hindi, and Punjabi can be selected at any stage of the experience. This makes the tool more relevant to local and regional users.

Action-oriented results. The score is not presented as a judgement. It is described as a starting point and connected directly to practical recommendations such as Google Pay for Business, Google Business Profile, Google Sheets, WhatsApp Business, and Canva Free.

Persistent submissions. The full-stack project stores the selected language, answers, category scores, total score, percentage, and timestamp in a database. This creates a foundation for future dashboards, analytics, and follow-up support.

Responsive interaction. The layout adapts to desktop and mobile screens, with accessible buttons, visible progress, selection states, and a clear next action.

5 Interface Evidence

The following screenshot shows the final landing screen of the tool. It demonstrates the main visual language, the three-language selector, the plain-language introduction, and the first action button.

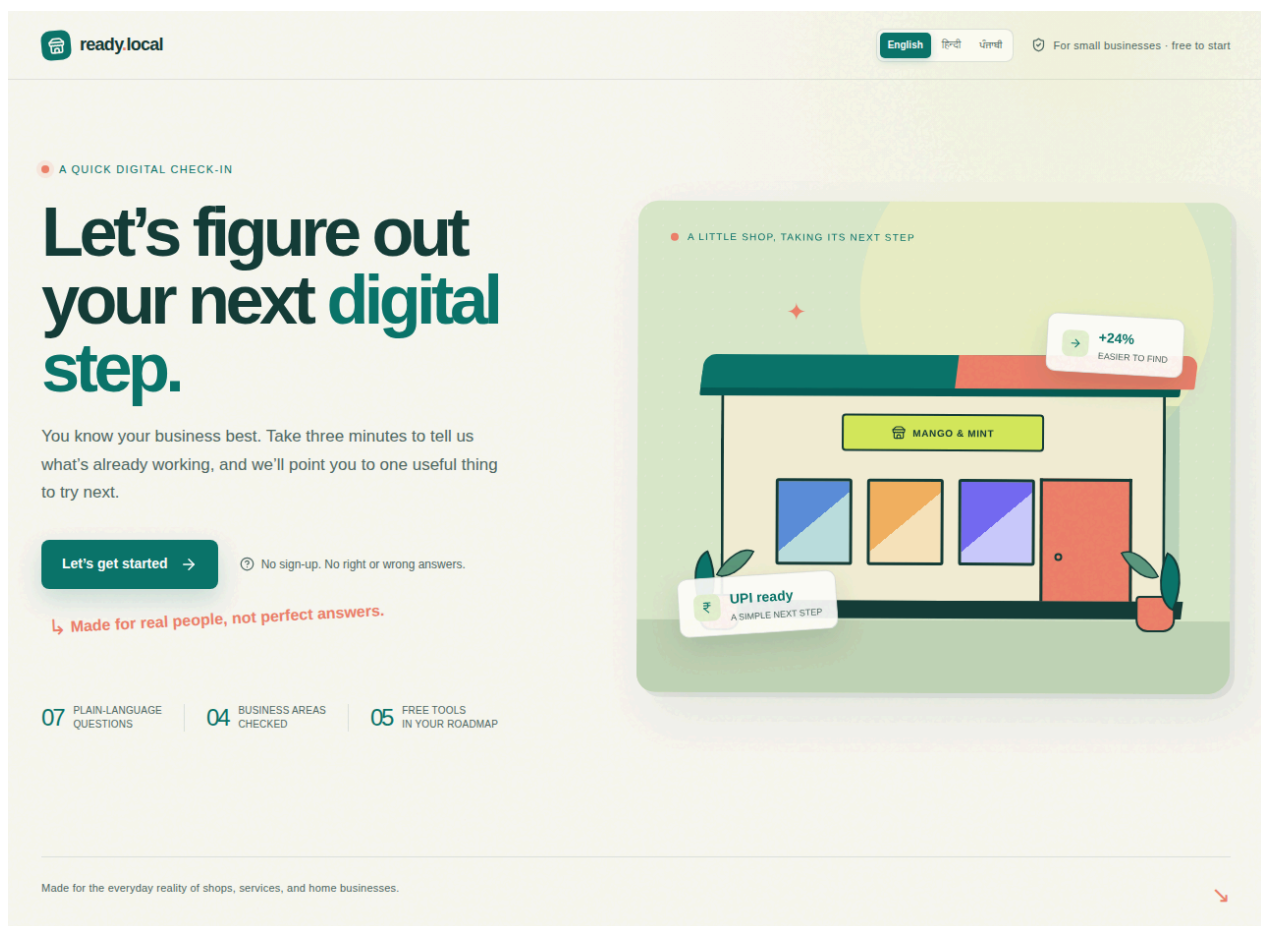


图 1 Final landing screen of the Digital Adoption Readiness Tool.

Live link: [Open the published website](#)

6 Technical Implementation

The frontend was built with React and TypeScript using a Vite-based web application structure. The interface is organised as a state-driven quiz flow with three primary states: introduction, quiz, and

results. The scoring logic maps each answer to a point value, calculates category percentages, and sorts recommendations according to the user's largest opportunity.

The database-backed version uses an Express server, tRPC procedures, Drizzle ORM, and a MySQL-compatible database. A `readinessSubmissions` table stores the completed check-in in a compact and extensible form. The public submission procedure validates the language, answers, scores, and percentage before writing the record.

The project was validated through the following checks:

Validation	Outcome
TypeScript check	Passed with no type errors.
Production build	Passed; frontend and server bundles generated.
Automated tests	3 tests passed across authentication and readiness submission behaviour.
Live browser flow	Completed quiz produced a localised roadmap and saved result feedback.
Database verification	A completed submission was retrieved from the database after the live flow.

7 Innovation and Value

The innovation in this project is not based on adding complexity. It lies in combining assessment, prioritisation, language access, and action planning in one small experience. Instead of giving a business owner a long list of digital possibilities, the tool creates a personal starting point from the user's own answers.

The design also treats confidence as part of adoption readiness. The final question asks how comfortable the user feels trying a new digital tool. This helps the roadmap remain proportionate: a person who is nervous about technology should receive a simpler first step, while a confident user can explore more advanced options.

8 Reflection and Future Scope

This project demonstrates how a design-thinking approach can turn a broad social and business problem into a focused digital intervention. The most important learning is that a technically correct product is not automatically a useful product. The wording, pace, visual tone, language choices, and size of the first action all influence whether a user feels able to continue.

Future improvements could include an owner dashboard for viewing anonymised trends, downloadable personalised reports, region-specific recommendations, optional business profiles, and a guided follow-up journey after the first recommendation is completed. The Netlify deployment currently hosts the public frontend; a production-ready release would connect the database-backed server through Netlify Functions or another secure backend host.

9 Conclusion

The Digital Adoption Readiness Tool is a practical, human-centred response to the challenge of digital adoption among small businesses. It uses a short check-in to understand the user's current position, transforms the result into a clear roadmap, and keeps the experience accessible through multilingual content and encouraging language.

As a Semester 1 project for **Design Thinking and Innovation**, it demonstrates problem framing, user-centred ideation, prototyping, interface design, implementation, testing, and reflection. The project shows that innovation can be meaningful when it reduces confusion and helps a real person take one useful next step.

10 Appendix: Project Details

Item	Details
Project title	Digital Adoption Readiness Tool
Subject	Design Thinking and Innovation
Student	Chiku
Programme	BE CSE AI & ML
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Academic level	Semester 1
Published website	digital-adoption-readiness-tool.netlify.app

Thank you

One useful digital step at a time.